The Mackey Formula as Monad Decomposition

Notes

Abstract

We give a detailed account of the Mackey formula through two worked examples: S_n inside S_{n+1} (two double cosets, the Heisenberg relation) and $\mathrm{GL}_n(\mathbb{F}_q)$ inside $\mathrm{GL}_{n+1}(\mathbb{F}_q)$ ($q^2 + 2q - 2$ double cosets, new field-arithmetic channels). We explain the monad $\mathrm{Ind} \circ \mathrm{Res}$, whose Mackey decomposition recovers the Hecke algebra as Kleisli endomorphisms, and connect to Sabidussi coset graphs and Fomin's local rules.

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1 The Mackey formula

1.1 Double cosets

Let G be a finite group and $H \leqslant G$ a subgroup. The set of *double cosets* $H\backslash G/H$ partitions G into disjoint subsets $HgH = \{hgk : h, k \in H\}$.

The key geometric idea: every conjugate gHg^{-1} is another copy of H inside G. The double coset HgH measures *how this conjugate copy sits relative to the original*, up to the symmetry of H itself. Two elements g, g' give the same double coset if and only if the "relative position" of gHg^{-1} and H, as seen by H, is the same.

Each double coset HgH determines:

- the *intersection subgroup* $H \cap gHg^{-1} \leqslant H$ (how much the two copies overlap),
- the *index* $[H : H \cap gHg^{-1}] = |HgH|/|H|$ (how "far apart" the copies are).

1.2 Twisted modules

Let M be a representation of H. For each $g \in G$, define M_g to be the vector space M with the H-action twisted by g: an element $h \in H$ acts by

$$h \cdot_g m \;=\; (g^{-1}hg) \cdot m.$$

This makes sense because $g^{-1}hg$ lies in $g^{-1}Hg \cong H$, so we are just composing the original action with the automorphism $h \mapsto g^{-1}hg$ of H. When $g \in H$, this is an inner automorphism and $M_g \cong M$.

From M_g we can build an H-module N_g by restricting to the intersection $H \cap gHg^{-1}$ and inducing back:

$$N_g \;:=\; \mathrm{Ind}_{H\cap gHg^{-1}}^{H}\; \mathrm{Res}_{H\cap gHg^{-1}}^{gHg^{-1}}\; M_g.$$

Lemma 1.1. *If $g' = h_1 g h_2$ for some $h_1, h_2 \in H$, then $N_{g'} \cong N_g$ as H-modules. So N_g depends only on the double coset HgH.*

Proof. We check the two factors h_1 and h_2 separately.

Right multiplication: $g' = gh_2$. Then $g'Hg'^{-1} = g(h_2 H h_2^{-1})g^{-1} = gHg^{-1}$ (since $h_2 \in H$ normalises H). So the conjugate subgroup is unchanged, the intersection $H \cap g'Hg'^{-1} = H \cap gHg^{-1}$ is unchanged, and the twisted module $M_{g'} = M_{gh_2}$ has action $h \cdot_{g'} m = (h_2^{-1} g^{-1} h g h_2) \cdot m = h_2^{-1} \cdot (h \cdot_g (h_2 \cdot m))$. This is M_g conjugated by the action of $h_2 \in H$ on M, which gives an isomorphic module after induction.

Left multiplication: $g' = h_1 g$. Then $g'Hg'^{-1} = h_1(gHg^{-1})h_1^{-1}$, and $H \cap g'Hg'^{-1} = H \cap h_1(gHg^{-1})h_1^{-1} = h_1(H \cap gHg^{-1})h_1^{-1}$ (since $h_1 \in H$). Conjugation by h_1 gives an isomorphism

$$\mathrm{Ind}_{H \cap gHg^{-1}}^{H}(-) \xrightarrow{\sim} \mathrm{Ind}_{h_1(H \cap gHg^{-1})h_1^{-1}}^{H}(-)$$

(induction from conjugate subgroups gives isomorphic results), and $M_{h_1 g}$ corresponds to M_g under this conjugation. So $N_{h_1 g} \cong N_g$. □

1.3 The Mackey decomposition

Theorem 1.2 (Mackey)**.** *Let $H \leqslant G$ and let M be a representation of H. Then*

$$\mathrm{Res}_H^G \ \mathrm{Ind}_H^G(M) \ \cong \bigoplus_{HgH \in H\backslash G/H} N_g \ = \bigoplus_{HgH \in H\backslash G/H} \mathrm{Ind}_{H \cap gHg^{-1}}^{H} \ \mathrm{Res}_{H \cap gHg^{-1}}^{gHg^{-1}} \ M_g. \qquad (1)$$

Proof. Recall that the induced representation is

$$\mathrm{Ind}_H^G(M) = k[G] \otimes_{k[H]} M,$$

the tensor product of M with the group algebra of G over the group algebra of H. The tensor is over $k[H]$, meaning we impose $xh \otimes m = x \otimes h \cdot m$ for all $h \in H$. As a k-vector space, choosing one representative x from each left coset $Hx \in H\backslash G$, we have

$$\mathrm{Ind}_H^G(M) = \bigoplus_{Hx \in H\backslash G} x \otimes M.$$

An element $g \in G$ acts by left multiplication on the first factor:

$$g \cdot (x \otimes m) = (gx) \otimes m. \qquad (2)$$

If gx lies in a different coset Hy, write $gx = hy$ for a unique $h \in H$, and then $(gx) \otimes m = hy \otimes m = y \otimes h \cdot m$ (using the $k[H]$-tensor relation). So g moves the element from the Hx-summand to the Hgx-summand, and the H-part of the coset change acts on m.

Example 1.3. G acts on the set of left cosets $G/H = \{aH : a \in G\}$ by left multiplication: $g \cdot (aH) = (ga)H$. This is well-defined because $gaH = ga'H$ whenever $aH = a'H$. Linearising this action (taking the vector space with basis $\{e_{aH}\}$ and extending $g \cdot e_{aH} = e_{gaH}$ linearly) gives the *permutation representation* $k[G/H]$. The induced representation of the trivial module is exactly this: $\mathrm{Ind}_H^G(\mathbf{1}_H) \cong k[G/H]$.

Now consider $\mathrm{Res}^G_H \mathrm{Ind}^G_H(M)$: we have the G-module $\mathrm{Ind}^G_H(M) = \bigoplus_{Hx \in H\backslash G} x \otimes M$, and we restrict the action to $H \leqslant G$. The G-action permuted the summands transitively, but H only permutes them among themselves according to how H acts on $H\backslash G$. The coset space $H\backslash G$ decomposes into orbits under the left H-action, and these orbits are exactly the double cosets: $H\backslash G = \bigsqcup_{HgH}(H\backslash HgH)$. Within a single double coset HgH, the right cosets $\{Hx : x \in HgH\}$ form the set $H\backslash HgH$, which is in bijection with $H/(H \cap gHg^{-1})$ via $Hx \leftrightarrow H \cap gHg^{-1} \cdot y$ (where $x = gy$ for appropriate y).

So the H-module $\mathrm{Res}^G_H \mathrm{Ind}^G_H(M)$ decomposes as

$$\mathrm{Res}^G_H \mathrm{Ind}^G_H(M) = \bigoplus_{HgH \in H\backslash G/H} \underbrace{\bigoplus_{Hx \in H\backslash HgH} x \otimes M}_{\text{contribution of the double coset } HgH} .$$

It remains to identify each inner sum as $\mathrm{Ind}^H_{H\cap gHg^{-1}} \mathrm{Res}^{gHg^{-1}}_{H\cap gHg^{-1}} g \cdot M$.

Fix a double coset HgH. Any element $x \in HgH$ can be written as $x = hgh'$ for some $h, h' \in H$. The right cosets of H inside HgH are $\{Hgh' : h' \in H\}$, but $Hgh'_1 = Hgh'_2$ if and only if $h'_2(h'_1)^{-1} \in g^{-1}Hg \cap H = g^{-1}(H \cap gHg^{-1})g$. So the right cosets are indexed by $(H \cap gHg^{-1})\backslash H$ (from the right action of H on gH).

Thus the inner sum, as an H-module under the left action, is

$$\bigoplus_{Hx \in H\backslash HgH} x \otimes M \;\cong\; k[H] \otimes_{k[H\cap gHg^{-1}]} (g \otimes M),$$

where $H \cap gHg^{-1}$ acts on $g \otimes M$ by $\gamma \cdot (g \otimes m) = g \otimes (g^{-1}\gamma g) \cdot m$ for $\gamma \in H \cap gHg^{-1}$ (noting that $g^{-1}\gamma g \in H$). This is precisely

$$\mathrm{Ind}^H_{H\cap gHg^{-1}} \mathrm{Res}^{gHg^{-1}}_{H\cap gHg^{-1}} g \cdot M. \qquad \square$$

Remark 1.4. More generally, for two subgroups $H, K \leqslant G$, the same argument gives

$$\mathrm{Res}^G_K \mathrm{Ind}^G_H(M) \;\cong \bigoplus_{KgH \in K\backslash G/H} \mathrm{Ind}^K_{K\cap gHg^{-1}} \mathrm{Res}^{gHg^{-1}}_{K\cap gHg^{-1}} g \cdot M.$$

The proof is identical, decomposing $K\backslash G$ into (K, H)-double cosets instead of (H, H)-double cosets.

Each double coset contributes one summand. The summand for HgH routes M through a *bottleneck*: restrict to $H \cap gHg^{-1}$ (the overlap between the original and conjugate copies of H), then induce back. The smaller the intersection, the tighter the bottleneck, and the more the representation is "reshuffled."

- The **identity coset** $HeH = H$ gives intersection $H \cap H = H$ (no bottleneck). Its summand is M itself.
- Every **non-trivial coset** gives a proper subgroup $H \cap gHg^{-1} \lneq H$. The representation must squeeze through this smaller group and then expand back.

Remark 1.5. Notice that G barely appears on the right-hand side of (1). The sum is over the double cosets $H\backslash G/H$ (which depend on G), and the conjugation $g \cdot M$ uses an element $g \in G$. But every operation on representations—the restriction, the induction, the subgroups involved—takes place entirely within H and its subgroups. The ambient group G sets the *stage* (determining which double cosets exist and which intersections arise), but does not *act* in the decomposition. The formula reduces a question about G-representations to bookkeeping inside H.

1.4 The Hecke algebra

Definition 1.6. Let $\mathcal{H}(G,H)$ be the space of H-bi-invariant functions on G:

$$\mathcal{H}(G,H) = \{f : G \to k \mid f(h_1 g h_2) = f(g)\ \forall\, h_1, h_2 \in H\},$$

with the *convolution product* $(f_1 * f_2)(g) = \sum_{x \in G} f_1(x)\, f_2(x^{-1}g)$.

Since H-bi-invariant functions are constant on double cosets, the characteristic functions $\{e_{HgH}\}$ form a basis, so $\dim \mathcal{H}(G,H) = |H\backslash G/H|$.

Proposition 1.7. *$\mathcal{H}(G,H)$ is a unital associative algebra under convolution.*

Proof. We must check three things.

Closure. If f_1, f_2 are H-bi-invariant, then so is $f_1 * f_2$. For any $h_1, h_2 \in H$:

$$\begin{aligned}
(f_1 * f_2)(h_1 g h_2) &= \sum_{x \in G} f_1(x)\, f_2(x^{-1} h_1 g h_2) \\
&= \sum_{x \in G} f_1(x)\, f_2(x^{-1} h_1 g) \qquad \text{(right } H\text{-invariance of } f_2\text{)} \\
&= \sum_{y \in G} f_1(y h_1)\, f_2(y^{-1} g) \qquad \text{(substituting } y = x h_1^{-1}\text{)} \\
&= \sum_{y \in G} f_1(y)\, f_2(y^{-1} g) \qquad \text{(right } H\text{-invariance of } f_1\text{)} \\
&= (f_1 * f_2)(g).
\end{aligned}$$

Associativity. This follows from associativity of multiplication in G:

$$((f_1 * f_2) * f_3)(g) = \sum_{x,y} f_1(x)\, f_2(x^{-1}y)\, f_3(y^{-1}g) = (f_1 * (f_2 * f_3))(g).$$

Unit. The function $e_H = \frac{1}{|H|}\mathbf{1}_H$ (the characteristic function of H, scaled by $1/|H|$) is the identity: $(e_H * f)(g) = \frac{1}{|H|}\sum_{h \in H} f(h^{-1}g) = f(g)$ by H-bi-invariance of f. □

We call $\mathcal{H}(G,H)$ the *Hecke algebra* of the pair (G,H).

By the Frobenius–Mackey argument (Frobenius reciprocity applied twice):

$$\mathcal{H}(G,H) \cong \mathrm{End}_G\big(\mathrm{Ind}_H^G(\mathbf{1}_H)\big). \tag{3}$$

Each basis element e_{HgH} is a G-endomorphism of the permutation representation $k[G/H]$. In other words, every double coset defines an operator on G/H, and these operators form an algebra under composition.

2 Example 1: S_n inside S_{n+1}

2.1 The embedding

Embed S_n into S_{n+1} as the subgroup fixing the element $n{+}1$:

$$H = \mathrm{Stab}(n{+}1) = \{\sigma \in S_{n+1} : \sigma(n{+}1) = n{+}1\} \;\cong\; S_n.$$

The coset space S_{n+1}/S_n has $[S_{n+1} : S_n] = n+1$ elements, one for each possible image of $n{+}1$: the coset σH consists of all permutations that send $n{+}1$ to $\sigma(n{+}1)$.

2.2 Conjugate copies of S_n

A conjugate $gHg^{-1} = g\,\mathrm{Stab}(n{+}1)\,g^{-1} = \mathrm{Stab}(g(n{+}1))$ is the subgroup fixing $g(n{+}1)$. There are $n{+}1$ such subgroups:

$$\mathrm{Stab}(1),\ \mathrm{Stab}(2),\ \ldots,\ \mathrm{Stab}(n),\ \mathrm{Stab}(n{+}1) = H.$$

All are copies of S_n inside S_{n+1}, and they are all conjugate.

2.3 Why there are exactly 2 double cosets

From $H = \mathrm{Stab}(n{+}1)$'s perspective, the $n{+}1$ conjugate copies fall into exactly **two** classes:

1. $\mathrm{Stab}(n{+}1)$ itself: the original copy. (1 subgroup.)
2. $\mathrm{Stab}(k)$ for $k \in \{1, \ldots, n\}$: a copy that fixes a *different* point. These all look the same to H, because H acts transitively on $\{1, \ldots, n\}$—for any $k, k' \in \{1, \ldots, n\}$, there exists $h \in H$ with $h(k) = k'$, and conjugating $\mathrm{Stab}(k)$ by h gives $\mathrm{Stab}(k')$. (n subgroups, all equivalent under H-conjugacy.)

This is the H-absorption principle: $n{+}1$ conjugate copies, but H can shuffle n of them into each other, leaving 2 orbits.

The two double cosets are:

- **Identity coset** $HeH = H$: the set of permutations fixing $n{+}1$. Size $|H| = n!$.
- **Non-trivial coset** $H(k,\, n{+}1)H$: the set of all permutations that *move* $n{+}1$. Size $|S_{n+1}| - |H| = (n+1)! - n! = n \cdot n!$. (This is the same double coset for every $k \in \{1, \ldots, n\}$.)

Dimension check: $n! + n \cdot n! = n!(1+n) = (n+1)! = |S_{n+1}|$. ✓

2.4 The intersection (the bottleneck)

For the non-trivial double coset, pick representative $g = (k,\, n{+}1)$. Then $gHg^{-1} = \mathrm{Stab}(k)$, and the intersection is:

$$H \cap gHg^{-1} = \mathrm{Stab}(n{+}1) \cap \mathrm{Stab}(k) = \{\sigma \in S_{n+1} : \sigma(n{+}1) = n{+}1 \text{ and } \sigma(k) = k\} \cong S_{n-1}.$$

This is the subgroup that fixes *both* points $n{+}1$ and k, acting on the remaining $n-1$ elements.

Remark 2.1. The intersection S_{n-1} depends on which representative k we choose, but different choices give conjugate copies of S_{n-1} inside S_n, and the Mackey summand is the same up to isomorphism.

2.5 The Mackey formula for $S_n \subset S_{n+1}$

Combining the two double cosets, we get for any S_n-module M:

$$\boxed{\mathrm{Res}^{S_{n+1}}_{S_n}\ \mathrm{Ind}^{S_{n+1}}_{S_n}(M)\ \cong\ M\ \oplus\ \mathrm{Ind}^{S_n}_{S_{n-1}}\ \mathrm{Res}^{S_n}_{S_{n-1}}(M).} \tag{4}$$

The identity coset ($g = e$): $H \cap eHe^{-1} = H$. The bottleneck is H itself—nothing is lost. The summand is $\mathrm{Ind}_H^H \mathrm{Res}_H^H(M) = M$. The representation passes through unchanged.

The non-trivial coset ($g = (k,\, n{+}1)$): $H \cap gHg^{-1} = S_{n-1}$. The bottleneck is one level smaller. The summand routes M through S_{n-1}:

$$M \xrightarrow{\text{Res}} \mathrm{Res}^{S_n}_{S_{n-1}}(M) \xrightarrow{\text{Ind}} \mathrm{Ind}^{S_n}_{S_{n-1}} \mathrm{Res}^{S_n}_{S_{n-1}}(M).$$

The representation "forgets" one level of structure (restriction to S_{n-1}) and then "relearns" it (induction back to S_n), possibly ending up in a different irreducible component.

Remark 2.2 (The Heisenberg relation). Writing $U = \mathrm{Ind}_{S_n}^{S_{n+1}}$ and $D = \mathrm{Res}_{S_n}^{S_{n+1}}$, equation (4) says

$$DU = \mathrm{Id} + UD,$$

i.e., $DU - UD = \mathrm{Id}$. This is the *Heisenberg commutation relation*. The identity coset contributes the Id and the non-trivial coset contributes UD.

Summing over all n with generating function weights, this becomes the Pieri operator identity $\Omega^*[Xu]\,\Omega[Xv] = \frac{1}{1-uv}\,\Omega[Xv]\,\Omega^*[Xu]$, where the factor $\frac{1}{1-uv}$ packages the identity coset contributions across all levels.

2.6 What happens to an irreducible: walks on Young's lattice

Let $M = S^\mu$ be the irreducible S_n-module labelled by a partition $\mu \vdash n$. By the branching rule for the chain $S_n \supset S_{n-1}$:

Restriction (remove one box): $\mathrm{Res}^{S_n}_{S_{n-1}}(S^\mu) = \bigoplus_\nu S^\nu$, summing over all partitions ν obtained from μ by removing one box (i.e., μ/ν is a single box).

Induction (add one box): $\mathrm{Ind}^{S_n}_{S_{n-1}}(S^\nu) = \bigoplus_\lambda S^\lambda$, summing over all partitions $\lambda \vdash n$ obtained from ν by adding one box.

Combining: the non-trivial Mackey summand is

$$\mathrm{Ind}^{S_n}_{S_{n-1}}\, \mathrm{Res}^{S_n}_{S_{n-1}}(S^\mu) = \bigoplus_{\nu} \bigoplus_{\lambda \supset \nu} S^\lambda, \tag{5}$$

where ν ranges over μ-minus-a-box and λ ranges over ν-plus-a-box. This is a **two-step walk on Young's lattice**: down one level and back up.

The complete Mackey decomposition is therefore:

$$\mathrm{Res} \circ \mathrm{Ind}(S^\mu) \;=\; S^\mu \;\oplus \bigoplus_{\substack{\nu \subset \mu \\ |\nu| = n-1}} \bigoplus_{\substack{\lambda \supset \nu \\ |\lambda| = n}} S^\lambda.$$

2.7 Worked example: S_2 inside S_3

Take $n = 2$, so $H = S_2$ (fixing 3) inside $G = S_3$. The two irreducibles of S_2 are $S^{(2)}$ (trivial) and $S^{(1,1)}$ (sign).

Case $M = S^{(2)}$: Remove a box from (2): the only option is $\nu = (1)$. Add a box to (1): get (2) or (1, 1). So

$$\mathrm{Res} \circ \mathrm{Ind}(S^{(2)}) = S^{(2)} \oplus \left(S^{(2)} \oplus S^{(1,1)}\right) = (S^{(2)})^{\oplus 2} \oplus S^{(1,1)}.$$

Dimension check: $\dim = 2 + 1 = 3 = [S_3 : S_2] \cdot \dim S^{(2)} = 3 \cdot 1$. ✓

Case $M = S^{(1,1)}$: Remove a box from $(1,1)$: the only option is $\nu = (1)$. Add a box to (1): get (2) or $(1,1)$. So

$$\mathrm{Res} \circ \mathrm{Ind}(S^{(1,1)}) = S^{(1,1)} \oplus \left(S^{(2)} \oplus S^{(1,1)}\right) = S^{(2)} \oplus (S^{(1,1)})^{\oplus 2}.$$

Dimension check: $1 + 2 = 3$. ✓

Remark 2.3. In both cases, $\mathrm{Res} \circ \mathrm{Ind}(M)$ has dimension $[G : H] \cdot \dim M = 3 \cdot 1 = 3$. The identity coset contributes dimension 1 (returning M) and the non-trivial coset contributes dimension 2 (the index $[S_2 : S_1] = 2$ times $\dim M = 1$).

2.8 The general case: $S_m \times S_n$ inside S_{m+n}

Embed $H = S_m \times S_n$ into $G = S_{m+n}$ as the subgroup preserving the partition $\{1, \ldots, m\} \mid \{m+1, \ldots, m+n\}$. A conjugate gHg^{-1} preserves a *different* partition of $\{1, \ldots, m+n\}$ into blocks of size m and n.

From H's perspective, what matters is: how much do the two partitions overlap? If the new block of size m shares exactly $m - r$ elements with the old block of size m, then r elements "crossed over."

Proposition 2.4. *The double cosets $(S_m \times S_n) \backslash S_{m+n} / (S_m \times S_n)$ are indexed by $r = 0, 1, \ldots, \min(m, n)$, giving $\min(m, n) + 1$ double cosets. The intersection for the double coset indexed by r is*

$$H \cap gHg^{-1} \;\cong\; (S_{m-r} \times S_r) \times (S_r \times S_{n-r}) \;\cong\; S_{m-r} \times S_r \times S_r \times S_{n-r},$$

where r elements from each block have crossed into the other.

The case $S_n \subset S_{n+1}$ is the special case $m = n$, $n = 1$ (or rather, $H = S_n \times S_1$, but S_1 is trivial). Then $\min(n, 1) + 1 = 2$ double cosets, recovering our earlier result.

3 Example 2: $\mathrm{GL}_n(\mathbb{F}_q)$ and the Borel subgroup

The right analogue of $S_n \subset S_{n+1}$ is not $\mathrm{GL}_n(\mathbb{F}_q) \subset \mathrm{GL}_{n+1}(\mathbb{F}_q)$ (which has too many double cosets for a clean story) but rather $(\mathrm{GL}_n(\mathbb{F}_q),\, B_n(\mathbb{F}_q))$, where B_n is the Borel subgroup of upper triangular matrices.

3.1 The Borel subgroup

Let $G = \mathrm{GL}_n(\mathbb{F}_q)$ and let $B = B_n(\mathbb{F}_q)$ be the subgroup of upper triangular matrices:

$$B = \left\{ \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix} : a_{ii} \in \mathbb{F}_q^*,\ a_{ij} \in \mathbb{F}_q \text{ for } i < j \right\}.$$

The order is $|B| = (q-1)^n \cdot q^{\binom{n}{2}}$.

3.2 The flag variety

B is the stabiliser of the *standard flag*

$$F_\bullet : \quad \{0\} \subset \langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \cdots \subset \langle e_1, \ldots, e_n \rangle = \mathbb{F}_q^n.$$

The coset space G/B is the *complete flag variety*: the set of all complete flags $\{0\} = V_0 \subset V_1 \subset \cdots \subset V_n = \mathbb{F}_q^n$ with $\dim V_i = i$.

For $n = 2$: a flag is just a line in $\mathbb{F}_q^2$, so $G/B = \mathbb{P}^1(\mathbb{F}_q)$ with $q + 1$ points.

For $n = 3$: a flag is an incident line-plane pair (ℓ, π) with $\ell \subset \pi \subset \mathbb{F}_q^3$.

3.3 The Bruhat decomposition

The Weyl group of GL_n is $W = S_n$, realised as permutation matrices. The key structural result:

Theorem 3.1 (Bruhat)**.** $\mathrm{GL}_n(\mathbb{F}_q) = \bigsqcup_{w \in S_n} BwB$. *The double cosets $B\backslash \mathrm{GL}_n / B$ are indexed by the symmetric group S_n.*

Proof for $n = 2$. $W = \{e, s\}$ where $s = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Any $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$: if $c = 0$ then $g \in B = BeB$; if $c \neq 0$ then $g \in BsB$ (use row and column operations from B to reduce g to s). So $\mathrm{GL}_2 = B \sqcup BsB$. □

Compare with the S_n case: $S_n \backslash S_{n+1} / S_n$ had 2 double cosets (the point stabiliser absorbs everything). Here $B\backslash \mathrm{GL}_n / B$ has $n!$ double cosets—one per permutation. The Borel is "too small" to absorb conjugate copies, so the full Weyl group is visible.

3.4 The bottlenecks

For each $w \in S_n$, the intersection $B \cap wBw^{-1}$ is the bottleneck for the double coset BwB. Its size depends on w:

- $w = e$: $B \cap B = B$. Full overlap, no bottleneck.
- $w = s_i$ (simple reflection): $B \cap s_i B s_i^{-1}$ has index q in B—a thin bottleneck. The missing "q directions" correspond to the q lines in a local $\mathbb{P}^1$.
- $w = w_0$ (longest element): $B \cap w_0 B w_0^{-1} = T$ (the torus, diagonal matrices). The tightest bottleneck—all the upper-triangular structure is lost.

In general, $[B : B \cap wBw^{-1}] = q^{\ell(w)}$ where $\ell(w)$ is the Bruhat length (number of simple reflections in a reduced expression for w). So longer elements have tighter bottlenecks.

3.5 The Mackey formula for (GL_n, B)

Applying Theorem 1.2 to the pair $(G, H) = (\mathrm{GL}_n, B)$:

$$\mathrm{Res}^G_B \ \mathrm{Ind}^G_B(M) \ \cong \ \bigoplus_{w \in S_n} \mathrm{Ind}^B_{B \cap wBw^{-1}} \ \mathrm{Res}^{wBw^{-1}}_{B \cap wBw^{-1}} \ M_w. \tag{6}$$

This has $n!$ summands, one per permutation. Each summand routes M through the bottleneck $B \cap wBw^{-1}$, which has codimension $\ell(w)$ in B.

3.6 The quadratic relation from $\mathbb{P}^1(\mathbb{F}_q)$

We now derive the Hecke algebra relations from this Mackey decomposition. Start with $n = 2$, where $B\backslash \mathrm{GL}_2/B$ has two elements and $\mathcal{H}(\mathrm{GL}_2, B)$ is 2-dimensional with basis $\{T_e, T_s\}$.

The permutation representation $k[G/B] = k[\mathbb{P}^1(\mathbb{F}_q)]$ has $q+1$ basis vectors δ_L, one per line L in $\mathbb{F}_q^2$. The operator T_s (the double coset BsB) sends each point to the sum of all other points:

$$T_s \cdot \delta_L = \sum_{L' \neq L} \delta_{L'}. \tag{7}$$

This is because BsB consists of matrices sending $\langle e_1 \rangle$ to a different line, and B acts transitively on the q lines other than $\langle e_1 \rangle$.

Theorem 3.2. $(T_s - q)(T_s + 1) = 0$.

Proof. Apply T_s twice to δ_L:

$$T_s^2 \cdot \delta_L = \sum_{L' \neq L} \sum_{L'' \neq L'} \delta_{L''}.$$

Count how often each L'' appears:

- $L'' = L$: each of the q choices of $L' \neq L$ contributes δ_L once (since $L \neq L'$). Total: $q \cdot \delta_L$.
- $L'' \neq L$: among the q choices of L' (all $\neq L$), exactly one is excluded ($L' = L''$), leaving $q - 1$. Total: $(q-1) \cdot \delta_{L''}$.

So $T_s^2 = q \cdot T_e + (q-1) \cdot T_s$, which factors as $(T_s - q)(T_s + 1) = 0$. □

Remark 3.3. The eigenvalues q and -1 correspond to: $T_s = q$ on the trivial representation (the constant function: T_s sends it to q times itself), and $T_s = -1$ on the q-dimensional complement (functions summing to zero: $T_s = J - I$ acts as $-I$ on $\ker J$).

3.7 The braid relation from flags in $\mathbb{F}_q^3$

For $n = 3$, the generators T_{s_1} and T_{s_2} act on flags (ℓ, π) by:

- T_{s_1}: replace the line ℓ by another line in π (sum over q choices).
- T_{s_2}: replace the plane π by another plane through ℓ (sum over q choices).

Each is a "local T_s" on a copy of $\mathbb{P}^1(\mathbb{F}_q)$ (the lines in a plane, or the planes through a line), so each satisfies $(T_{s_i} - q)(T_{s_i} + 1) = 0$.

Theorem 3.4. $T_{s_1} T_{s_2} T_{s_1} = T_{s_2} T_{s_1} T_{s_2}$.

Proof. Both sides produce a sum over flags at Bruhat distance 3 from (ℓ, π). $T_{s_1} T_{s_2} T_{s_1}$ traces the path $(\ell, \pi) \to (\ell', \pi) \to (\ell', \pi') \to (\ell'', \pi')$ (change line, change plane, change line). $T_{s_2} T_{s_1} T_{s_2}$ traces $(\ell, \pi) \to (\ell, \pi') \to (\ell', \pi') \to (\ell', \pi'')$ (change plane, change line, change plane). Both equal T_{w_0} (the basis element for the longest element of S_3), because $s_1 s_2 s_1 = s_2 s_1 s_2 = w_0$ and both products are length-additive: $\ell(w_0) = 3 = 1 + 1 + 1$. □

3.8 The isomorphism

Definition 3.5. The *Iwahori–Hecke algebra* $H_n(q)$ is the associative algebra generated by $T_1, \dots, T_{n-1}$ subject to:

$$(T_i - q)(T_i + 1) = 0 \qquad \text{for all } i, \tag{8}$$

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1} \qquad \text{for all } i, \tag{9}$$

$$T_i T_j = T_j T_i \qquad \text{for } |i - j| > 1. \tag{10}$$

At $q = 1$: $(T_i - 1)(T_i + 1) = 0$ gives $T_i^2 = 1$, and $H_n(1) = \mathbb{C}[S_n]$.

Theorem 3.6 (Iwahori, 1964)**.** *$H_n(q) \cong \mathcal{H}(\mathrm{GL}_n(\mathbb{F}_q), B_n(\mathbb{F}_q))$, sending T_i to the characteristic function of Bs_iB. Both algebras have dimension $n!$.*

3.9 Comparison: S_n versus (GL_n, B)

	$(\boldsymbol{S_{n+1}, S_n})$	$(\mathbf{GL}_{\boldsymbol{n}}(\mathbb{F}_{\boldsymbol{q}}), \boldsymbol{B_n}(\mathbb{F}_{\boldsymbol{q}}))$
Double cosets	2	$n!$ (Bruhat)
Indexed by	{identity, non-trivial}	S_n (Weyl group)
Hecke algebra dim	2	$n!$
Bottleneck for w	S_{n-1}	$B \cap wBw^{-1}$, index $q^{\ell(w)}$
Quadratic relation	$T_s^2 = 1$ (involution)	$(T_s - q)(T_s + 1) = 0$
At $q = 1$	—	recovers $\mathbb{C}[S_n]$
Mackey gives	Heisenberg $DU - UD = I$	Iwahori–Hecke algebra $H_n(q)$
Centralises	$\mathfrak{gl}_N$ on $V^{\otimes n}$	$U_q(\mathfrak{gl}_N)$ on $V^{\otimes n}$

The parameter q counts the points of a "local $\mathbb{P}^1(\mathbb{F}_q)$" (lines in a plane, or planes through a line). The quadratic relation $(T_s - q)(T_s + 1) = 0$ encodes this geometry: T_s has eigenvalue q on the constant function and -1 on its complement. As $q \to 1$, $\mathbb{P}^1$ shrinks to two points, T_s becomes a transposition, and $H_n(q)$ degenerates to $\mathbb{C}[S_n]$.

4 The Hall algebra: q-deformed Heisenberg

For $S_n \subset S_{n+1}$, two double cosets gave the Heisenberg relation $DU - UD = \mathrm{Id}$ (Remark 2.2). What replaces this for $\mathrm{GL}_n(\mathbb{F}_q)$?

4.1 Finite modules and their extensions

Let $\mathbb{F}_q$ be a finite field. A *finite $\mathbb{F}_q[t]$-module* is a finite-dimensional $\mathbb{F}_q$-vector space V equipped with an $\mathbb{F}_q$-linear endomorphism $T : V \to V$. Equivalently, it is a finitely generated $\mathbb{F}_q[t]$-module that is killed by some power of t (we consider only nilpotent endomorphisms, for simplicity).

By the structure theorem for modules over a PID, every such module is isomorphic to

$$M_\lambda = \mathbb{F}_q[t]/(t^{\lambda_1}) \oplus \cdots \oplus \mathbb{F}_q[t]/(t^{\lambda_\ell}),$$

for a unique partition $\lambda = (\lambda_1 \geqslant \cdots \geqslant \lambda_\ell)$. The partition λ records the sizes of the Jordan blocks of T, and $|M_\lambda| = q^{|\lambda|}$.

Definition 4.1. For partitions λ, μ, ν, the *Hall number* $g^\lambda_{\mu\nu}(q)$ is the number of submodules $N \subseteq M_\lambda$ such that $N \cong M_\nu$ and $M_\lambda/N \cong M_\mu$:

$$g^\lambda_{\mu\nu}(q) = \big|\{N \leqslant M_\lambda : N \cong M_\nu,\ M_\lambda/N \cong M_\mu\}\big|.$$

The Hall numbers count *extensions*: the number of ways to build M_λ as an extension of M_μ by M_ν, up to the automorphisms already accounted for. Philip Hall (1959) showed that $g^\lambda_{\mu\nu}(q)$ is a polynomial in q with integer coefficients.

Example 4.2. $g^{(2)}_{(1)(1)}(q)$: how many submodules $N \cong M_{(1)} = \mathbb{F}_q$ of $M_{(2)} = \mathbb{F}_q[t]/(t^2)$ have $M_{(2)}/N \cong M_{(1)}$? The module $M_{(2)}$ has underlying space $\mathbb{F}_q^2$ with T acting by $T(a,b) = (b,0)$. The only T-invariant subspace isomorphic to $\mathbb{F}_q$ with the correct quotient is $\ker(T) = \{(a,0)\}$. So $g^{(2)}_{(1)(1)}(q) = 1$.

$g^{(1,1)}_{(1)(1)}(q)$: how many submodules $N \cong \mathbb{F}_q$ of $M_{(1,1)} = \mathbb{F}_q \oplus \mathbb{F}_q$ (where $T = 0$) have quotient $\cong \mathbb{F}_q$? Any one-dimensional subspace works, and there are $q+1$ of them (the lines in $\mathbb{F}_q^2$). But each is isomorphic to $\mathbb{F}_q$ with $T = 0$, so $g^{(1,1)}_{(1)(1)}(q) = q+1$.

4.2 Definition of the Hall algebra

Definition 4.3. The *Hall algebra* $\mathcal{A}_q$ is the $\mathbb{Z}$-module freely generated by symbols $[M_\lambda]$ for each partition λ, with multiplication

$$[M_\mu] \cdot [M_\nu] = \sum_\lambda g^\lambda_{\mu\nu}(q)\ [M_\lambda]. \tag{11}$$

Proposition 4.4. *The Hall algebra $\mathcal{A}_q$ is a unital associative algebra.*

Proof. Associativity: $([M_\mu]\cdot[M_\nu])\cdot[M_\rho]$ counts the number of chains $0 \subset N_1 \subset N_2 \subset M_\lambda$ with $N_1 \cong M_\rho$, $N_2/N_1 \cong M_\nu$, $M_\lambda/N_2 \cong M_\mu$. This equals $[M_\mu]\cdot([M_\nu]\cdot[M_\rho])$, which counts the same chains (a filtration of M_λ with graded pieces M_ρ, M_ν, M_μ is the same data regardless of whether we group the first two or last two).

Unit: $[M_\varnothing] = [0]$, corresponding to the empty partition, satisfies $[0]\cdot[M_\lambda] = [M_\lambda]$ since the only submodule isomorphic to M_λ of M_λ with zero quotient is M_λ itself. □

4.3 The simplest case: multiplying two copies of $[M_{(1)}]$

The element $[M_{(1)}]$ corresponds to the one-dimensional module $\mathbb{F}_q$ (with $T = 0$). Computing $[M_{(1)}]^2$:

$$[M_{(1)}]\cdot[M_{(1)}] = g^{(2)}_{(1)(1)}(q)\,[M_{(2)}] + g^{(1,1)}_{(1)(1)}(q)\,[M_{(1,1)}].$$

From Example 4.2: $g^{(2)}_{(1)(1)} = 1$ and $g^{(1,1)}_{(1)(1)} = q+1$. So

$$[M_{(1)}]^2 = [M_{(2)}] + (q+1)\,[M_{(1,1)}]. \tag{12}$$

For comparison, in the ring of symmetric functions (the $q \to \infty$ / $t \to 0$ limit), we have $h_1^2 = h_2 + e_2$, i.e., the coefficient of e_2 is just 1, not $q+1$. The factor $q+1$ counts the lines in $\mathbb{F}_q^2$—this is the finite field enrichment that deforms the symmetric function product.

4.4 The Hall algebra is a deformation of symmetric functions

There is an isomorphism (the *characteristic map*)

$$\mathcal{A}_q \xrightarrow{\sim} \Lambda \otimes_{\mathbb{Z}} \mathbb{Z}[q],$$

from the Hall algebra to the ring of symmetric functions with coefficients extended to $\mathbb{Z}[q]$, sending $[M_\lambda]$ to a scalar multiple of the Hall–Littlewood polynomial $P_\lambda(x;\, q^{-1})$. Under this isomorphism, the Hall product becomes a q-deformation of the product of symmetric functions.

In particular, defining $u_n = [M_{(1^n)}]$ (the class of $\mathbb{F}_q^n$ with $T = 0$) and $e_n = [M_{(n)}]$ (the class of $\mathbb{F}_q[t]/(t^n)$), we get q-deformed elementary and complete symmetric functions.

4.5 The commutation relation

We now derive the q-Heisenberg relation. Define creation and annihilation operators on $\mathcal{A}_q$ by:

$$U(v) = \text{multiplication by } \sum_{n \geqslant 0} [M_{(1^n)}]\, v^n \quad (\text{“add a semisimple module”}),$$

$$D(u) = \text{adjoint operator (with respect to the Hall pairing)}.$$

The Hall pairing is $\langle [M_\mu], [M_\nu] \rangle = \delta_{\mu\nu}\, |\mathrm{Aut}(M_\mu)|$, where $|\mathrm{Aut}(M_\mu)|$ is the number of automorphisms of M_μ.

The key computation: $D(u)\,U(v)$ counts pairs (submodule, quotient) while $U(v)\,D(u)$ counts the same pairs in the opposite order. The discrepancy between the two orderings is controlled by the number of automorphisms, which introduces factors of q.

Theorem 4.5 (Jing, 1991)**.** *The vertex operators for the Hall algebra satisfy*

$$D(u)\,U(v) = \frac{1 - uv/q}{1 - uv}\; U(v)\,D(u). \tag{13}$$

Proof sketch. At the level of modes, the underlying Lie algebra relation is

$$[a_m, a_n^\dagger] = m\,(1 - q^{-|m|})\,\delta_{m+n,0},$$

where $a_m, a_n^\dagger$ are the annihilation and creation modes. At $q = \infty$ (i.e., $q^{-|m|} = 0$), this is the standard Heisenberg relation $[a_m, a_n^\dagger] = m\,\delta_{m+n,0}$.

The factor $(1 - q^{-|m|})$ arises because the automorphism group $\mathrm{Aut}(M_\lambda)$ has order divisible by powers of q: for instance $|\mathrm{Aut}(\mathbb{F}_q^n)| = |\mathrm{GL}_n(\mathbb{F}_q)| = \prod_{i=0}^{n-1}(q^n - q^i)$. When you commute creation past annihilation, the ratio of automorphism groups for the two orderings (submodule first vs. quotient first) produces the factor $(1 - q^{-|m|})$.

Exponentiating the mode relation gives the generating function form (13). □

4.6 Comparison with the Heisenberg relation

Writing $t = 1/q$:

Symmetric groups ($t = 0$)		**Hall algebra** ($t = 1/q$)
$D(u)\,U(v)$ $\frac{1}{1-uv}\,U(v)\,D(u)$	$=$	$D(u)\,U(v) = \frac{1-tuv}{1-uv}\,U(v)\,D(u)$
$[a_m, a_n^\dagger] = m\,\delta_{m+n,0}$		$[a_m, a_n^\dagger] = m(1-t^{\lvert m\rvert})\,\delta_{m+n,0}$
Schur functions s_λ		Hall–Littlewood $P_\lambda(x;\,t)$
Hall numbers $= 0$ or 1		Hall numbers $\in \mathbb{Z}[q]$
$\lvert\mathrm{Aut}(M_\lambda)\rvert = 1$		$\lvert\mathrm{Aut}(M_\lambda)\rvert =$ polynomial in q

The extra factor $(1-tuv)$ in the numerator can be traced to the automorphism groups of $\mathbb{F}_q[t]$-modules. In the symmetric group case ($q \to \infty$, $t \to 0$), these automorphism groups become trivial and the factor disappears.

4.7 Limiting cases

- $t = 0$ ($q \to \infty$): the factor becomes $\frac{1}{1-uv}$, recovering the Heisenberg relation. The symmetric group case is the "generic" case where the finite field structure contributes nothing.
- $t = 1$ ($q = 1$, formal limit): the factor becomes $\frac{1-uv}{1-uv} = 1$, so $DU = UD$ and creation and annihilation commute. This is the "maximally abelian" limit.
- $t = 1/q$ for a prime power q: the actual $\mathrm{GL}_n(\mathbb{F}_q)$ case. The commutation factor $\frac{1-uv/q}{1-uv}$ interpolates between the Heisenberg algebra and the abelian limit.

Remark 4.6. The Hall–Littlewood polynomials $P_\lambda(x;\,t)$ interpolate between Schur functions ($t = 0$) and monomial symmetric functions ($t = 1$). At $t = 1/q$, they encode the representation theory of $\mathrm{GL}_n(\mathbb{F}_q)$ in the same way that Schur functions encode S_n.

5 The monad and its decomposition

5.1 $\mathrm{Ind} \circ \mathrm{Res}$ as a monad

For finite groups, induction and restriction participate in two adjunctions:

$$\mathrm{Ind}_H^G \;\dashv\; \mathrm{Res}_H^G \qquad \text{(Frobenius reciprocity)}, \tag{14}$$

$$\mathrm{Res}_H^G \;\dashv\; \mathrm{Ind}_H^G \qquad \text{(since Ind} \cong \text{Coind for finite groups)}. \tag{15}$$

From adjunction (15), $T = \mathrm{Ind}_H^G \circ \mathrm{Res}_H^G : \mathcal{R}\mathrm{ep}(G) \to \mathcal{R}\mathrm{ep}(G)$ is a **monad**. The Mackey formula decomposes this monad into a direct sum of functors, one per double coset.

5.2 The Hecke algebra and the monad

The Hecke algebra $\mathcal{H}(G, H)$ (Definition 1.6) has dimension $\lvert H\backslash G/H\rvert$: for $S_n \subset S_{n+1}$ this is 2, for $\mathrm{GL}_1(\mathbb{F}_3) \subset \mathrm{GL}_2(\mathbb{F}_3)$ this is 13. By the isomorphism $\mathcal{H}(G, H) \cong \mathrm{End}_G(\mathrm{Ind}_H^G(\mathbf{1}_H))$

from (3), each basis element e_{HgH} is a G-endomorphism of the permutation representation—and these are exactly the adjacency operators of the Sabidussi coset graphs $\mathrm{Sab}(G, H, g)$.

The monad perspective explains *why* the Hecke algebra has a multiplication: the monad $T = \mathrm{Ind} \circ \mathrm{Res}$ is composable, and the monad multiplication $T^2 \to T$ (applying two Ind∘Res cycles and merging them) gives the convolution product on double coset operators.

5.3 Eilenberg–Moore algebras (T-algebras)

A T-algebra is a G-representation V with a structure map $\alpha : T(V) = \mathrm{Ind}(\mathrm{Res}(V)) \to V$ satisfying unit and associativity axioms. Via the Mackey decomposition, α decomposes as one projection α_{HgH} per double coset.

For finite groups in characteristic zero, the Barr–Beck theorem gives $\mathcal{R}\mathrm{ep}(G) \simeq T\text{-Alg}$: a G-representation is the same as an H-representation with coherent descent data (one projection per double coset, satisfying compatibility conditions).

5.4 The Kleisli category

The Kleisli category $\mathrm{Kl}(T)$ has the same objects as $\mathcal{R}\mathrm{ep}(G)$ but morphisms

$$\mathrm{Hom}_{\mathrm{Kl}(T)}(V, W) = \mathrm{Hom}_G(V,\, \mathrm{Ind}\ \mathrm{Res}(W)) \cong \mathrm{Hom}_H(\mathrm{Res}(V),\, \mathrm{Res}(W)).$$

A Kleisli morphism is an H-equivariant map between restrictions: $\mathcal{R}\mathrm{ep}(G)$ with morphisms relaxed to H-equivariance.

The Hecke algebra appears as the Kleisli endomorphism algebra of the trivial representation under the monad $T' = \mathrm{Res} \circ \mathrm{Ind}$ on $\mathcal{R}\mathrm{ep}(H)$:

$$\mathrm{End}_{\mathrm{Kl}(T')}(\mathbf{1}_H) \cong \mathrm{Hom}_G(\mathrm{Ind}(\mathbf{1}_H),\, \mathrm{Ind}(\mathbf{1}_H)) = \mathcal{H}(G, H).$$

Each double coset = one Kleisli endomorphism = one Sabidussi graph = one Mackey summand.

6 Connection to local rules and Schur–Weyl duality

6.1 The Schur–Weyl double cosets

For $G = S_{m+n}$ and $H = S_m \times S_n$, the $\min(m, n) + 1$ double cosets (indexed by the crossing parameter r) control the Mackey decomposition of $\mathrm{Ind} \circ \mathrm{Res}$ for representations of the symmetric group.

The Pieri operator $\Omega[Xv]$ (generating function for induction along $S_n \hookrightarrow S_{n+1}$) and its dual $\Omega^*[Xu]$ satisfy

$$\Omega^*[Xu]\ \Omega[Xv] = \frac{1}{1 - uv}\ \Omega[Xv]\ \Omega^*[Xu].$$

This is the Mackey formula for the pair $(S_{n+1},\, S_n)$, with the factor $\frac{1}{1-uv} = \sum_{m \geqslant 0} (uv)^m$ arising from the non-trivial double coset.

6.2 Local rules as Mackey steps

Fomin's local rule for growth diagrams computes this Mackey decomposition one cell at a time: the integer m in the local rule $\mathfrak{U}_{\alpha,\beta}(m,\mu)=\lambda$ is the identity-coset contribution, and the partition data tracks the routing through the non-trivial bottleneck.

6.3 The categorical framework

Davydov–Molev [1] define Schur–Weyl categories $\mathcal{C}(A_*)$ from multiplicative sequences of algebras. The Hecke algebra appears in two guises:

1. $\mathcal{H}(G,H)$: the double-coset Hecke algebra (this paper).
2. $\mathrm{H}_n(q)$: the Iwahori–Hecke algebra, which equals $\mathcal{H}(\mathrm{GL}_n(\mathbb{F}_q),\, B_n(\mathbb{F}_q))$ for the Borel subgroup B_n.

The passage from $k[S_n]$ to $\mathrm{H}_n(q)$ in Davydov–Molev categorifies the passage from the symmetric group pair (2 double cosets, clean bottleneck) to the $\mathrm{GL}_n(\mathbb{F}_q)$ pair (many double cosets, field-arithmetic channels).

7 Summary

Concept	$\boldsymbol{S_n \subset S_{n+1}}$	$\boldsymbol{\mathrm{GL}_1(\mathbb{F}_q) \subset \mathrm{GL}_2(\mathbb{F}_q)}$
Double cosets	2 (fix same point, or different)	q^2+2q-2 (field invariants)
Hecke algebra dim	2	q^2+2q-2
Bottleneck	S_{n-1} (one level down)	trivial (maximal reshuffling)
Central channels	none	$q-2$ (from $Z(\mathrm{GL}_2)$)
Mackey formula	$M \oplus \mathrm{Ind}\,\mathrm{Res}(M)$	$(q-1)$ copies of M plus reshuffled terms
Heisenberg relation	$UD = I + DU$	more complex commutation
Sabidussi graphs	1 non-trivial graph (valency n)	q^2+2q-3 non-trivial graphs

The unifying principle: double cosets count how many essentially different ways a conjugate copy of H can sit inside G, up to the symmetry of H itself. For S_n, H's transitivity absorbs almost all the copies, leaving just 2 classes. For $\mathrm{GL}_n(\mathbb{F}_q)$, the field structure creates additional invariants (d, the normalised A-entry) that H cannot absorb, producing polynomially many classes in q. The Mackey formula decomposes the monad $\mathrm{Ind}\circ\mathrm{Res}$ along these classes, the Hecke algebra records their convolution structure, and the local rules / Sabidussi graphs are individual channels of the decomposition.

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